MANIPAL
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MANIPAL SCHOOL OF INFORMATION SCIENCES

(A Constituent unit of MAHE, Manipal)

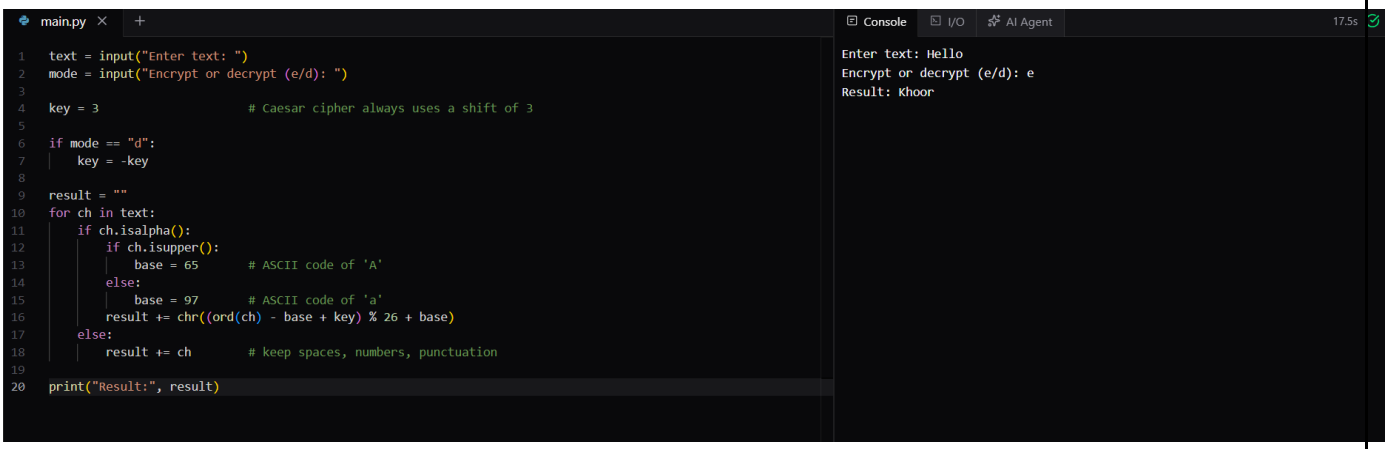
Cryptology – Lab_5

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1. **Write Python code to encrypt/decrypt:**
    I. **Caesar Cipher:**



Caesar cipher is just the shift cipher with the key fixed at 3, so the script doesn't ask you for a key. Everything else works the same way.

Output 1:

Enter text: Hello

Encrypt or decrypt (e/d): e

Result: Khoor

Output 2:

Enter text: Khoor

Encrypt or decrypt (e/d): e

Result: Hello

II. **Shift Cipher:**



- ord(ch) - base turns a letter into a number from 0 to 25 (A = 0, B = 1, and so on).
- + key shifts the number.
- % 26 wraps around, so Z shifted by 1 goes back to A.
- chr(... + base) turns the number back into a letter.

- For decryption, the key is made negative, so it shifts backwards.

Output 1:
Enter text: Cryptology
Enter key: 3
Encrypt or decrypt (e/d): e
Result: Fubswrorjb
Output 2:
Enter text: Khoor
Enter key: 3
Encrypt or decrypt (e/d): d
Result: Hello

III. ROT 13:



ROT13 is a shift of 13, and since the alphabet has 26 letters, shifting by 13 twice brings you back to the start. So there's no encrypt/decrypt choice.
Output 1:
Enter text: Welcome
Enter key: 3
Encrypt or decrypt (e/d): e
Result: Zhofrph

Output 2:
Enter text: Khoor
Enter key: 3
Encrypt or decrypt (e/d): d
Result: Hello

IV. Atbash Cipher:

main.py

```python
text = input("Enter text: ")

result = ""
for ch in text:
    if ch.isalpha():
        if ch.isupper():
            base = 65       # ASCII code of 'A'
        else:
            base = 97       # ASCII code of 'a'
        result += chr((25 - (ord(ch) - base)) + base)
    else:
        result += ch        # keep spaces, numbers, punctuation

print("Result:", result)
```

Atbash flips the alphabet, so A becomes Z, B becomes Y, C becomes X, and so on. The code turns each letter into a number (A = 0 to Z = 25) and replaces it with 25 - number

Output 1:

Enter text: ATTACKATONCE

Result: ZGGZXPZGLMXV

Output 2:

Enter text: svool

Result: hello

V. Affine Cipher:

main.py

```python
text = input("Enter text: ")
mode = input("Encrypt or decrypt (e/d): ")
a = int(input("Enter key a (1,3,5,7,9,11,15,17,19,21,23,25): "))
b = int(input("Enter key b: "))

# a must have no common factor with 26
if a not in [1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25]:
    print("Invalid a! Choose from the list.")
    exit()

if mode == "d":
    a_inv = pow(a, -1, 26)      # multiplicative inverse of a mod 26

result = ""
for ch in text:
    if ch.isalpha():
        if ch.isupper():
            base = 65           # ASCII code of 'A'
        else:
            base = 97           # ASCII code of 'a'
        x = ord(ch) - base      # letter -> number (A=0 ... Z=25)
        if mode == "e":
            y = (a * x + b) % 26
        else:
            y = (a_inv * (x - b)) % 26
        result += chr(y + base)
    else:
        result += ch            # keep spaces, numbers, punctuation

print("Result:", result)
```

- Encryption: $y = (a \cdot x + b) \bmod 26$
- Decryption: $x = a^{-1} \cdot (y - b) \bmod 26$
- a^{-1} is the number that gives 1 when multiplied by a and reduced mod 26. For a = 9, it is 3, because $9 \times 3 = 27 \equiv 1$.
- a must be in that list because, if it shares a factor with 26, two letters would encrypt to the same letter and decryption would be impossible.

Output 1:

Enter text: remainrebel
Encrypt or decrypt (e/d): e
Enter key a (1,3,5,7,9,11,15,17,19,21,23,25): 3
Enter key b: 5
Result: erpfdserirm

Output 2:

Enter text: svool
Encrypt or decrypt (e/d): d
Enter key a (1,3,5,7,9,11,15,17,19,21,23,25): 3
Enter key b: 9
Result: detts